

RENANCE

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EEE 321 — Past Questions & Materials
by Renance

Course: EEE 321
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200, 300

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Paper 1 of 3 — EEE 321 - 2024/2025 - Second Semester - Circuit Theory II

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

B.ENG EXAMINATION: 2nd Semester

SESSION: 2024/2025

DATE: 25/09/2025



COURSE CODE: EEE 321

COURSE TITLE: Circuit Theory II

TIME ALLOWED: 3Hours

INSTRUCTION: ANSWER FIVE QUESTIONS, LEAVE YOUR ANSWERS IN FOUR DECIMAL PLACE

Q1 a(i) Using figure 1.1, determine the Z , g and h parameters (14marks)

Q2 Using figure 2.1, find the following:

- (i) Open circuit impedance Z_{1oc} and Z_{2oc} (6marks)
- (ii) Short circuit impedance Z_{1sc} and Z_{2sc} (6marks)
- (iii) Image impedance Z_{m1} and Z_{m2} (2marks)

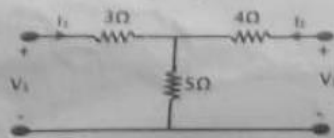


Fig. 1.1

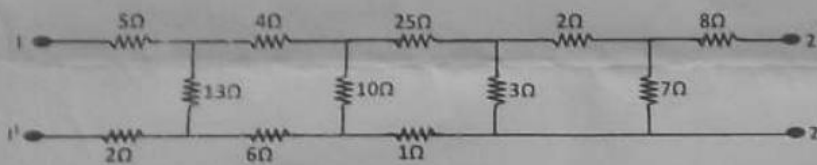


Fig. 2.1

Q3. a) Given that $Z_{1oc} = 280 - j40 \Omega$ and $Z_{1sc} = 172 + j14 \Omega$

- (i) The characteristic impedance (Z_0) (5.5marks)
- (ii) The Image Transfer coefficient (8.5marks)

4. a(i) Draw the labeled diagram of a prototype terminated constant K Band Stop Filter on the prototype T and π -section. (6marks)

b) Fig. 4.1 is a constant K filter Network; determine the values of the following:

- (i) Filter design impedance (R_0) (2marks)
- (ii) The cut-off frequency (f_c) (2marks)
- (iii) The attenuation (α) in dB of the filter at 60 KHz and 83 KHz. (4marks)

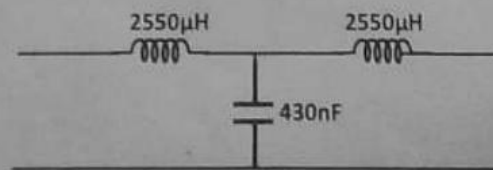


Fig. 4.1

$$G = 7.811 \angle -3.575$$

$$-8.130$$

$$172.5688 \angle 0.0812$$

$$Z_{1oc} Z_{1sc} = 48720 - 2960j$$

$$Z_{0c} = 488.01$$

- Q5. a(i) Design, draw and label the element values of a T-section constant K Band Pass filter of design impedance $0.125 \text{ k}\Omega$ and cut-off frequencies of 0.0016 GHz and 2.5 MHz . (8marks)
 b(i) Draw the label diagram of a prototype terminated constant K Band pass filter Based on the T and π -section prototype. (6marks)

- Q6. a) An R-L driving point impedance function is given by the expression

$$Z(s) = \frac{s(s+2)(s+5)}{(s+1)(s+4)}$$

(8marks)

(6marks)

- (i) Foster I and II form
 (ii) Cauer I and II form

- Q7. a) An R-C driving point impedance function is given by the expression

$$Z(s) = \frac{3(s+2)(s+4)}{s(s+3)}$$

(8marks)

(6marks)

- (i) Foster I and II form
 (ii) Cauer I and II form

$Z = 251.842 \text{ } \angle -0.1419$
 $Z_{in} = 112.947 \text{ } \angle 0.0512$

GOOD LUCK

Paper 2 of 3 — EEE 321 CIRCUIT THEORY II

Level 300 • Second Semester • PastQuestion

COLLEGE OF TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

B.ENG EXAMINATION: 2nd Semester

SESSION: 2023/2024

COURSE CODE: EEE 321

= COURSE TITLE: Circuit Theory II

DATE: 11/01/2024 ogc. TIME ALLOWED: 3Hours

INSTRUCTION: ANSWER FIVE QUESTIONS, LEAVE YOUR ANSWERS IN FOUR DECIMAL PLACE

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Q1. a(i)) Using figure 1.1, determine the Z, g and h parameters (14marks)

Q2 Using figure 2.1, find the following:

(i) Open circuit impedance Z_{ioc} and Z_{20c} (6marks)

(ii) Short circuit impedance Z_{isc} and Z_{esc} (6marks)

(iii) Image impedance, Z_{m2} Sex Lee (2marks)

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Fig. 1.1

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03. a) Given that $Z_{1oc} = 280 - j40 \Omega$ and $Z_{ise} = 172 + j14 \Omega$

(i) The characteristic impedance (Z_0) (5.4marks)

(ji) The Image Transfer coefficient . (8.5marks)

Q4. a(i) Draw the labeled diagram of a prototype terminated constant K Band Step Filter on the prototype T and tr-section. (6marks)

'b) Fig. 4.1 is a constant K filter Network: determine the values of the following:

(i) Filter design impedance $\{R_s\}$ (2marks)

(ii) The cut-off frequency (f_c) (2marks)

; fiche attenuation (a) in dB of the filter at 69 KHz and 83 KHz. (4marks)

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Fig. 4.4

QS. af) Design, draw and label the element values of a T-section constant K Band Pass filter of design impedance 0.125 k Ω and cut-off frequencies of 0.0016 GHz and

2.5 MHz. (8marks)

b(i) Draw the label diagram of a prototype terminated constant K Band pass filter Based on the T and re-section prototype. (6marks)

QG. a) An R-L driving point impedance function is given by the expression

$Z(s) = \frac{s^2 + 2}{s^2 + 5}$

Ze 'stints yay' Synthesize the impedance in

(i) Foster I and II form (8marks)

(ii) Gauer I and II form (6marks)

Q7. a) An R-C driving point impedance function is given by the expression

$Z(s) = \frac{s^2 + 544}{s^2 + 1}$

Ze 'say' Synthesize the impedance in

(i) Foster I and II form (8marks)

(ii) Cauer I and II form (6marks)

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Paper 3 of 3 — EEE 321 Circuit Theory II

Level 300 • Second Semester • PastQuestion

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B.ENG EXAMINATION: 2nd Semester
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COURSE TITLE: Circuit Theory II
TIME ALLOWED: 3Hours

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Q1. a(i) Using figure 1.1, determine the Z, g and h parameters (14marks)

Q2. Using Figure 2.1, find the following:

- Open circuit impedance Z_{1oc} and Z_{2oc}
- Short circuit impedance Z_{1sc} and Z_{2sc}
- Image impedance, Z_{m2}

Handwritten notes for Q2:
 $2 \cosh^{-1} \frac{f}{f_c}$
 $g = \frac{1}{4} + \frac{5}{10} + \frac{130}{23} + 7$
 $attenuation = 2 \cosh$
 $40 \times 4.2 \times 4.2$
 $20 \times 1.2 \times 20$
 $40 \times 1.2 \times 20$

Fig. 1.1

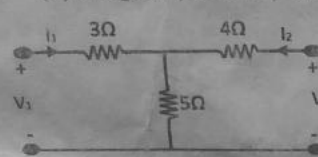
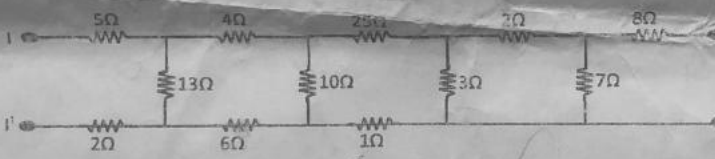


Fig. 2.1



Q3. a) Given that $Z_{1oc} = 280 - j40 \Omega$ and $Z_{1sc} = 172 + j14 \Omega$

- The characteristic impedance (Z_0)
- The Image Transfer coefficient

Handwritten notes for Q3:
 $R_0 = \sqrt{Z_0}$
 $where L/2 = \frac{2550 \times 10^{-6}}{2}$
 $= 1.25 \times 10^{-3}$
 $C = 430 \times 10^{-9}$

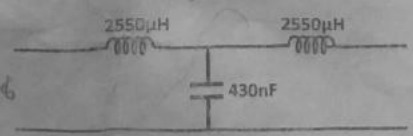
Q4. a(i) Draw the labeled diagram of a prototype terminated constant K Band Stop Filter on the prototype T and π -section. (6marks)

b) Fig. 4.1 is a constant K filter Network; determine the values of the following:

- Filter design impedance (R_0)
- The cut-off frequency (f_c)
- The attenuation (α) in dB of the filter at 60 KHz and 83 KHz.

Handwritten notes for Q4:
 $R_0 = \frac{1.125 \times 10^3}{430 \times 10^{-9}}$
 $= 51.15 \times 10^3$
 $f_c = \frac{1}{\sqrt{LC}}$
 $= \frac{1}{\sqrt{(1.125 \times 10^{-3})(430 \times 10^{-9})}}$
 $= 14472.4084$

Fig. 4.1



Handwritten calculations for Fig. 4.1:
 $\frac{25 \times 10}{22 \times 10} + \frac{26 \times 10}{26 \times 10}$
 $+ \frac{9 \times 10}{9 \times 10}$
 $\frac{15 \times 10}{15 \times 10} + \frac{3 \times 25}{3 \times 25} + \frac{1 \times 10}{1 \times 10} + \frac{10 \times 13}{10 \times 13}$
 $= 14472.4084$

6. a(i) Design, draw and label the element values of a **T-section** constant K Band Pass filter of design impedance **0.125 kΩ** and cut-off frequencies of **0.0016 GHz** and **2.5 MHz**. (8marks)
- b(i) Draw the label diagram of a prototype terminated constant K Band pass filter Based on the **T** and **π**-section prototype. (6marks)

Q6. a) An R-L driving point impedance function is given by the expression

$$Z(s) = \frac{s(s+2)(s+5)}{(s+1)(s+4)}$$

Synthesize the impedance in

- (i) Foster I and II form (8marks)
- (ii) Cauer I and II form (6marks)

Q7. a) An R-C driving point impedance function is given by the expression

$$Z(s) = \frac{3(s+2)(s+4)}{s(s+3)}$$

Synthesize the impedance in

- (i) Foster I and II form (8marks)
- (ii) Cauer I and II form (6marks)

$$Z_{in} = [(1+2s)/13] + (2s+1)/10 + (4s+6)/3 + 7$$

$$= \left[\frac{27}{12} + \frac{65}{9} + \frac{130}{22} + 7 \right] = 22.124 \Omega$$

GOOD LUCK